INCH-POUND

MIL-V-31001
25 February 1992
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

VEST, AMMUNITION CARRYING

This specification is approved by the U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of vest intended for carrying mortar shells.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8415

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SPECIFICATIONS

FEDERAL

A-A-203	Paper, Kraft, Untreated
V-T-285	Thread, Polyester
T-T-881	Twine, Cotton, Seine
DDD-L-20	Label; for Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-W-530	Webbing, Textile, Cotton, General Purpose. Natural or in Colors
MIL-B-543	Buckle, Tongueless, and Webstrap
MIL-H-639	Hook (For Belts, Straps and Equipage Items)
MIL-R-3390	Ring, Dee
MIL-H-9890	Hardware, Individual Load Carrying Equipment and Hardware, Miscellaneous
MIL-G-16491	Grommet, Metallic
MIL-L-35078	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for
MIL-C-43627	Cloth, Duck, Cotton, Plied Yarns, Fire, Water Weather and Mildew Resistant Treated, Lightdry Finish
MIL-S-43770/1	Snap, Hooks, General Specification For

STANDARDS

FEDERAL

FED-STD-751	Stitches, Seams and Stitching

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads
MIL-STD-731	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. Thefollowing other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DODISS Department of Defense Index of Specifications and Standards

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publications and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

U.S. Postal Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia,PA 19103-1187.)

NATIONAL MOTOR FREIGHT CLASSIFICATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), before production is commenced, finished vests shall be submitted, or made available to the contracting officer or his authorized representative for approval. The approval of the first article authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with all applicable provisions of the specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

3.2 Samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case, the specification shall govern.

3.3 Materials.

3.3.1 Basic Material. The basic material shall be cloth, cotton, duck, Olive Drab 7 shade, 17 ounce maximum per square yard; water, flame, and mildew resistant, conforming to MIL-C-43627.

3.3.2. Webbing - 3/4 inch. The 3/4 inch webbing for binding the edges of the vest shall be dyed olive drab 7, and shall conform to Type II, class 3 medium weight, 0.48 ounce minimum per linear yard of MIL-W-530.

3.3.2.1 Webbing - 1 inch. The 1 inch webbing for the side closure straps and shoulder reinforcement shall be dyed olive drab 7 and shall conform to Type III, class 3 heavyweight, 1.33 ounce minimum per linear yard of MIL-W-530.

3.3.3 Snap Hook. The snap hook shall be material 1, size 1, part No. CWBC-3G conforming to MIL-S-43770/1. The finish shall be black enamel.

3.3.4 Rings, Dee. The dee rings shall conform to configuration "K", 1 by 3/4 inch, class 1 or class 2 of MIL-R-3390 with a black enamel finish.

3.3.5 Grommet. The grommet shall conform to type I, class 7, size 2 of MIL-G-16491 with a black enamel finish.

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3.3.6 Twine. The cotton twine for securing the shells in the front and back vest pockets shall be No. 96 of T-T-881.

3.3.7 Hook and Eye. The hook and eye set shall conform to type I, class 2 of MIL-H-639. The hook has a double vertical loop 1-1/4 inches long, 5/8 inch wide at the hook and 3 1/2 inches wide at the base. The eye shall have a flat end to accommodate the hook and be of the proper size. The finish shall be black enamel.

3.3.8 Slide with depressed Center Bar (1 inch). The buckle shall conform to Type I, Style 2, construction (A) of MIL-B-543 with black enamel finish.

3.3.9 End Fastening (1 inch). The end fastening shall conform to Type VII, Class B of MIL-H-9890.

3.3.10 Thread. The thread shall be polyester conforming to type I, class 1, sub-class B of V-T-285 and in the following sizes:

For all stitching except binding and box "X" stitching	Size F
For box "X" stitching	Size FF
For prestitching and binding	Size E

The thread shall be dyed olive drab S-1 CA 66022, and shall show colorfastness to weathering equal to or better than the standard sample. When no standard sample is available, the thread shall show "good" colorfastness to weathering.

3.3.11 Identification label. The identification label shall conform to type VI, class 1 of DDD-L-20 and shall measure not less than 2-1/2 inches wide by 1-1/2 inches long. The content and format shall be as follows:

Nomenclature:
Contract Number:
Name of contractor:
NSN:

3.3.12 Markings. The marking of the letters "US" shall conform to type IV, class 9 of DDD-L-20 and be applied to the front pocket, centered in 1 inch high letters. Mark "front" and "back", respectively, on the inside of vest 2 1/2 inches from the bottom of vest and centered from each side. The marking shall be done in black with 1/2 inch letters.

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3.4 Design. The vest shall consist of front and back panels sewn together at the left shoulder and a hook and eye closure at the right shoulder. A pocket shall be sewn to the front and back panels for carrying a maximum of 3 shells with each pocket having a cotton twine for securing the load. Side closure straps shall be on each side with an adjustable buckle and snap fastener on each back panel and a Dee Ring closure on each front panel at the waist. The outer edges of the front and back panels including the neck opening shall be bound with 3/4 inch webbing (see 3.3.2).

3.4.1 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a maximum length of 1/4 inch.

3.4.2 Setting of grommets. Holes shall be prepunched to receive the grommets. Holes shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommet and washer shall be securely clinched without cutting or distorting the pocket material.

3.4.3 Dipping ends of Cotton Webbing. Unless otherwise specified, all ends of cotton webbing shall be dipped (50% Paraffin/50% Wax). Resulting seal will catch all yarn ends. Webbing ends shall be dipped prior to assembly for stitching.

3.5 Patterns. Patterns for the front and back panels contain marks for placing the pockets, side straps and twine reinforcement pieces, and shall be furnished by the Government. The pocket pattern has marks for grommet placement. The Government patterns shall not be altered in any way, except that additional notches for use during construction are allowed on the working patterns. The government patterns are to be used only as a guide for cutting the contractor's working patterns. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, servicebility or appearance requirements cited in the specification. The working patterns shall be identical to the Government patterns.

3.5.1 List of Pattern Parts. All parts shall be cut from basic fabric as follows:

Pattern Nomenclature	No. of cut parts
Front	1
Back	1
Pocket	2
Right Shoulder Reinforcement Place	1
Tab (Front and Back) Reinforcements	2
Hook and Eye Chape	2

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3.5.2 Cutting of Components. Pattern components shall have a ±1/8 inch tolerance.

3.5.3 Replacement of Defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3.1 shall be removed from production and replaced with non-defective and properly matched components.

3.6 Construction. The vests shall conform to the quality of product established by this specification.

3.6.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in Table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used.

3.6.1.1 Thread breaks and ends of seams. The ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitching, shall be backstitched not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches may be repaired by using 301 stitch type. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch.

3.6.1.2 Automatic Stitching. Automatic machines may be used to perform any of the stitch patterns provided the requirements for the stitch patterns, stitches per inch, and size and type of thread are met; and at least three or more tying, overlapping, or backstitches are used to secure the ends of stitches.

3.6.1.3 Abbreviations in Table of Operations. The abbreviations used in Table I are as follows:

Stch	-	Stitch
In	-	Inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
Smlr	-	Similar
Btn	-	Button
Btnhl	-	Buttonhole
Incl	-	Including

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TABLE I. Manufacturing requirements

NO	OPERATION	STCH TYPE	SEAM STCH TYPE	STCH IN	THREAD NDL	BOB LPR
1.	Cutting a. Cut front, back, shoulder reinforcement piece, tab reinforcements, and pockets in direction indicated by, by the directional lines on the pattern. When possible, top edge of pockets should be cut on the selvage. b. Cut the tabs from basic material (See 3.3.1) for securing the twine 2 inches by 5 inches c. Cut two pieces from from basic material (See 3.3.1) 5 inches by 7 inches for attaching shoulder closure hook and eye. d. Cut the 1 inch wide webbing (See 3.3.2.1) for the side closure straps as follows: two pieces 15 inches long for snap hooks and two pieces 3 inches long for securing the dee rings. Cut one piece 4 inches long for shoulder reinforcement. e. Cut two lengths of twine 31 inches long. Each end of the twine shall be overcast stitched for a distance of 1 inch.					
2.	Bind edges and sew label. Finished appearance. The binding shall be uniformly stitched and not be tight or full causing puckers, gathers, or pleats. The basic					

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TABLE I. Manufacturing requirements

NO	OPERATION	STCH TYPE	SEAM STCH TYPE	STCH IN	THREAD NDL	BOB LPR
	material and edges of binding shall be securely caught in the stitching.					
	a. Bind the outer raw edges of the entire front and back panels with the 3/4 inch webbing (See 3.3.2) with the stitching 1/16 inch from the edge of the webbing. Catch the top of the label on the inside back at center neck opening.	301	BSa-1	8-10	E	E
3.	Attach straps, reinforcement pieces and twine tabs.					
	a. Turn under the ends and edges of the twine tabs 3/8 inch and join one each to the front and back at the marks with a 1/8 inch seam, stitching around 4 sides of each (tab), securing center portion of twine with stitching. Overlap beginning not less than 1/2 inch on both ends of tabs.	301	LSd-1	8-10	F	F
	b. Turn under the ends of the 3 inch long webbing 3/8 inch and fold thru the dee ring and stitch at the marks on the front panel with a 3/4" Box "X" stitch, 1/8 inch from edges.	301	SSat-2	8-10	FF	FF
	c. Insert one end of the 15 inch webbing into end fastening. Insert the other end through the snap hook and feed through the buckle. Stitch dipped end of webbing to the back panel at marks, using 3/4 inch Box "X" stitch	301	EFb-1	8-10	FF	FF

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TABLE I. Manufacturing Operations

NO	OPERATION	STCH TYPE	SEAM STCH TYPE	STCH IN	THREAD NDL	BOB LPR
	d. Insert other end of webbing through the snap hook and feed through the buckle. Turn under the raw end 1/2 inch and stitch at the marks on the back panel using a "3/4" Box "X" stitch.	301	SSb-2	8-10	FF	FF
	e. Turn under the sides and ends of the 5 by 7 inch hook and eye reinforcing pieces and edge stitch with a 1/8 inch gauge.	301	EFa-1	8-10	E	E
	f. Join Reinforcement piece to under side of right shoulder; stitching 1/4 inch from the top of the re-inforcement piece. Stitch shall pass through reinforcement piece and shoulder when joining	301	SSa-1	8-10	F	F
	g. Put a reinforcing piece through the hook and a re-inforcing piece through the eye; folding each piece in half to form reinforce-ment. Abut ends on the under side of each re-enforcement and join using a 1 1/2 by 3 inch Box "X" stitch.	301	SSat-2	8-10	FF	FF
4.	Grommets. Finished appearance. The grommets shall be attached through the pre-punched holes without cutting or distorting the pocket.					
	a. Punch holes for grommets at the marks on the pockets and insert grommets from					

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TABLE I. Manufacturing Operations

NO	OPERATION	STCH TYPE	SEAM STCH TYPE	STCH IN	THREAD NDL	BOB LPR
	outside. Securely clinch over the backup washers on the under side of pocket.					
5.	Form and attach pockets.					
	a. Make mitered corner on pockets using 1/2 inch seam.					
	b. If top of pocket is not cut on selvage, turn under 1/2 inch and stitch with 1/8 inch gauge.	301	EFa-1	8-10	E	E
	c. Turn under the sides and bottom pocket edges 1/2 inch and stitch to the front and back with a double row of stitches using 1/4 inch gauge, 1/8 inch from folded edge.	301	SSb-1	8-10	E	E
6.	Join front to back.					
	a. Join front and back at left shoulder, joining 1/4 inch from the edge.	301	SSa-1	8-10	F	F
	b. Press Seam Allowance open flat.					
	c. Place 1 X 4 inch reinforcing webbing strip on underside of shoulder seam allowance. Stitch with two rows of stitching catching the front panel and two rows catching the rear panel. Outside row of stitching 1/8 inch from edge using 3/16 inch gauge. A single row of stitching will be across each end of the webbing. Stitching shall be on all four sides of reinforcement.	301	SSf-3	8-10	F	F

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TABLE I. Manufacturing Operations

NO	OPERATION	STCH TYPE	SEAM STCH TYPE	STCH IN	THREAD NDL	BOB LPR
7.	Markings					
	a. Place "US" marking on center of front pocket ensuring the letters are visible. b.Mark front" and "back", respectively, on the inside of the vest 2 1/2 inches from the bottom of the vest and centered from each side. The marking shall be done in black with 1/2 inch letter.					

3.7 Finished Measurements. The finished measurements of the vest shall conform to Table II.

Table II - Finished Measurements (in inches)

	Length 1/	Width 2/	Depth 3/
Front	22	15	-
Back	24	15	-
Pocket	13	12	3 1/2
Tolerance	± 1/2	± 1/2	± 1/2

1/ Measurement shall be taken with the vest opened (snaps not engaged) and laid smooth and flat. Measure from abutted shoulder seam at front to bottom of vest. Repeat process for back of vest, The pocket shall be measured from top to the bottom of the pocket at joining seam to body.

2/ Measurement shall be taken with vest opened (snaps not engaged) and laid smooth and flat. Measure front from finished side edge to the other finished side edge above and below pocket. Repeat process for back of vest. The pocket shall be measured on the front of the pocket from rounded corner to rounded corner.

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3/ Measurement shall be taken from the pocket/body joining seam to the rounded corner of the front edge of the pocket.

3.8 Workmanship. The finished vest shall conform to the quality of product established by this specification.

~~4. QUALITY ASSURANCE PROVISIONS~~

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are permitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.2 and 6.2), it shall be examined for the defects specified in 4.4.3, 4.4.4, and 4.4.5.

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4.4. Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and Material Certification. A certificate of compliance may be accepted as evidence that the slide (3.3.8), fastening (3.3.9), snap hook (3.3.3), rings, dee (3.3.4), grommet (3.3.5) and hook and eye (3.3.7) conform to the requirements specified.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing operation to determine whether cut lengths and cut parts, markings for location of components, and location of assembled component parts are in accordance with specified requirements. In addition, inspection shall be made to determine that prepunched holes for receiving grommets are smaller than the outside diameter of the hardware barrel. Whenever nonconformance is noted, correction shall be made to the parts affected and lot in process. Components which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end item shall be examinedfor the defects listed in Table III. The sample unit shall be one vest. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units,shall be 2.5 for major defects and 15 total (major and minorcombined) defects.

4.4.3.1 General defects. General defects shall be classified as follows:

TABLE III. End Item Visual Defects

Examine	Defect	Classification Major	Minor
Fabric	Hole, cut, tear, smash, broken or missing yarn, or open place clearly visible at normal inspection distance (approximately 3 feet)	101	
	Shade bar or abrasion mark		201

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TABLE III. End Item Visual Defects

Examine	Defect	Classification Major	Minor
Webbing	Any hole, cut, tear or smash	102	
	Not firmly and tightly woven	103	
	Edges frayed or scalloped	104	
	Multiple floats		202
	Abrasion mark, slub, or broken end or pick		203
Hardware (general)	Broken or malformed failing to serve intended purpose, corroded area, burr, or sharp edge	105	
	Finish omitted or not as specified	106	
	Area of partial or no finish		204
	Any required component improperly installed causing failure to serve intended purpose	107	
	Not assembled as specified	108	
	Size or type not as specified	109	
Twine, Cotton	Cut, chafed or abraded	110	
	Ends not overcast		205
Grommets	Clinched excessively tight cutting adjacent material	111	
	Insecurely clinched to degree that grommet may be detached from material	112	
	Washer installed on incorrect side of material		206
	Roll of grommet contains splits		207
Open seam	1/2-inch or less		208
	more than 1/2 inch	113	

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TABLE III. End Item Visual Defects

Examine	Defect	Classification Major	Minor
NOTE:	A seam shall be classified as an open seam when one or more stitches joining a seam ~~are broken, or when two or~~ more consecutive skipped or run-off stitches occur. On double stitched seams, a seam shall be considered open when either one or both sides of the seam are open.		
Raw edge (on edge required to be finished)	More than 1/2 inch when securely caught in stitching.		209
NOTE:	Raw edge not securely caught in stitching shall be classified as open seam.		
Run-off	(see open seam)		
Seam and Stitch tension	Wrong seam or stitch type	114	
	Loose, resulting in a loose bobbin or top thread		210
	Excessively tight, resulting in puckering of material		211
Stitches per inch	One or two stitches less than minimum specified		212
	Three or more stitches less than minimum specified	115	
	One or more stitches in excess of maximum specified		213
NOTE:	Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places or in turning corners,		

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TABLE III. End Item Visual Defects

Examine	Defect	Classification Major	Classification Minor
	shall be classified as follows: (a) Within the minor defect classification - no defect (b) Within the major defect classification - minor defect		
Stitching ends	Not secured as specified		214
Thread break skipped stitches, or run-offs	Not overstitched as specified		215
NOTE:	Thread breaks or two or more consecutive skipped or runoff stitches not over-stitched shall be class-ified as open seams.		
Rows of stitching	Any row missing except on box "X" stitching	116	
	On box "X" stitching: - one row of stitching omitted		216
	- two or more rows of stitching	117	
Components and assembly	Any component part omitted or not as specified or any operation omitted or not as specified (unless other-wise classified herein)	118	
	Needle chews	119	
	Any mend, darn, patch, splice or other unauthorized repair	120	
	Any material pleated or caught in stitch line where not specified		217
Cutting	Any component part not cut in conformance with directional lines indicated on patterns or not in accordance with specification	121	

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TABLE III. End Item Visual Defects

Examine	Defect	Classification Major	Minor
Pockets	Omitted or not located as specified	122	
Binding	~~Loosely applied but not exposing~~ raw edge of material		218
	Loosely applied exposing raw edge of material		219
Reinforcement webbing	Missing or not located as specified	123	
Cleanness	Grease, oil, dirt or ink stains clearly noticeable		220
	Thread ends not trimmed to 1/4-inch maximum length		221
Markings US, identification	Omitted, incorrect, illegible, misplaced, or size of characters not as specified		222

4.4.4 End item dimensional examination. The end items shall be examined for the defects listed in Table IV. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. The lot size shall be expressed in units of vests. The sample unit shall be one vest. The inspection level shall be S-3 and the AQL, expressed interms of defects per hundred units, shall be 6.5 for major defects and 15 total (major and minor combined) defects.

Table IV. End item dimensional defects

Examine	Defects	Classification Major	Minor
Dimensions (overall)	Smaller than nominal dimensions less applicable minus tolerance indicated on Figure 1, but not smaller than nominal dimensions less twice the applicable minus tolerance		201
	Smaller than nominal dimensions less twice the applicable minus tolerance.	101	

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TABLE IV End Item Dimensional Defects

Examine	Defects	Classification Major	Classification Minor
	Larger than nominal dimensions and applicable plus tolerance		202
Component and location dimensions	Not within specified tolerance		203
Box "X" stitching	Dimensions not as specified		204
Stitch margin and gauge	Not within specified tolerance		205
Grommets	Set off center or misplaced by more than 1/8 inch		206

4.4.5 Packaging examination. The fully packaged end item shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap. improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of vests per shipping container is more or less than required

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defects
Finished dimensions	Length, width or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect: illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Each vest shall have fasteners engaged and laid flat. Five (5) vests shall be stacked (alternating top and bottom) one on top of the other and tied together with cord, suitable for purpose intended, approximately 3 inches from each end of vest.

5.1.1 Level A. Fifteen vests (laid out flat), preserved as specified in 5.1, shall be neatly packed (alternately reversed top to bottom) one on top of the other within a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF,class weather-resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of each container shall approximate 22 inches in length, 18 inches in width and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, water-proofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.1.2 Level B. Fifteen vests (laid out flat), preserved as specified in 5.1, shall be neatly packed (alternately reversed top to bottom) one on top of the other within a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Inside dimensions of each shipping container shall

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approximate 22 inches in length, 18 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.1.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.1.3 Commercial. Vests shall be packed in accordance with ASTM D 3951.

5.2 Palletization. When specified (see 6.2), vests packed as specified in 5.1.1 and 5.1.2 shall be palletized on a 4-way entry pallet in accordance with load type 1a of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet used shall be approved by the contracting officer.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The vest is designed to carry three mortar shells each in the front and back pockets.

6.2 Acquisition Requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1).
d. Levels of Preservation and packing (see 5.1).
e. Type and class of unit load required (see 5.1.1).
f. When weather-resistant grade fiberboard containers are required for level B packing (see 5.1.2.1).
g. When palletization is required (see 5.2)

6.3 Standard Samples and Shade Samples. For information regarding the availability of sample vests and standard shades specified, address inquiry to the procuring activity issuing the invitation for bids.

6.4 Subject Term (key word) Listing.

Mortar
Shell

6.5 Drawings. This military specification supersedes D34951, D34794 and accompanying drawings.

6.6 Recycled materials. It is encouraged that recycled materials be used when practical as long as the materials meet the requirements of this specification.

Custodian:
Navy - MC

Review Activity:
Navy - MC
Army - GL

Preparing Activity

DLA-CT

Project No. 8415-0851

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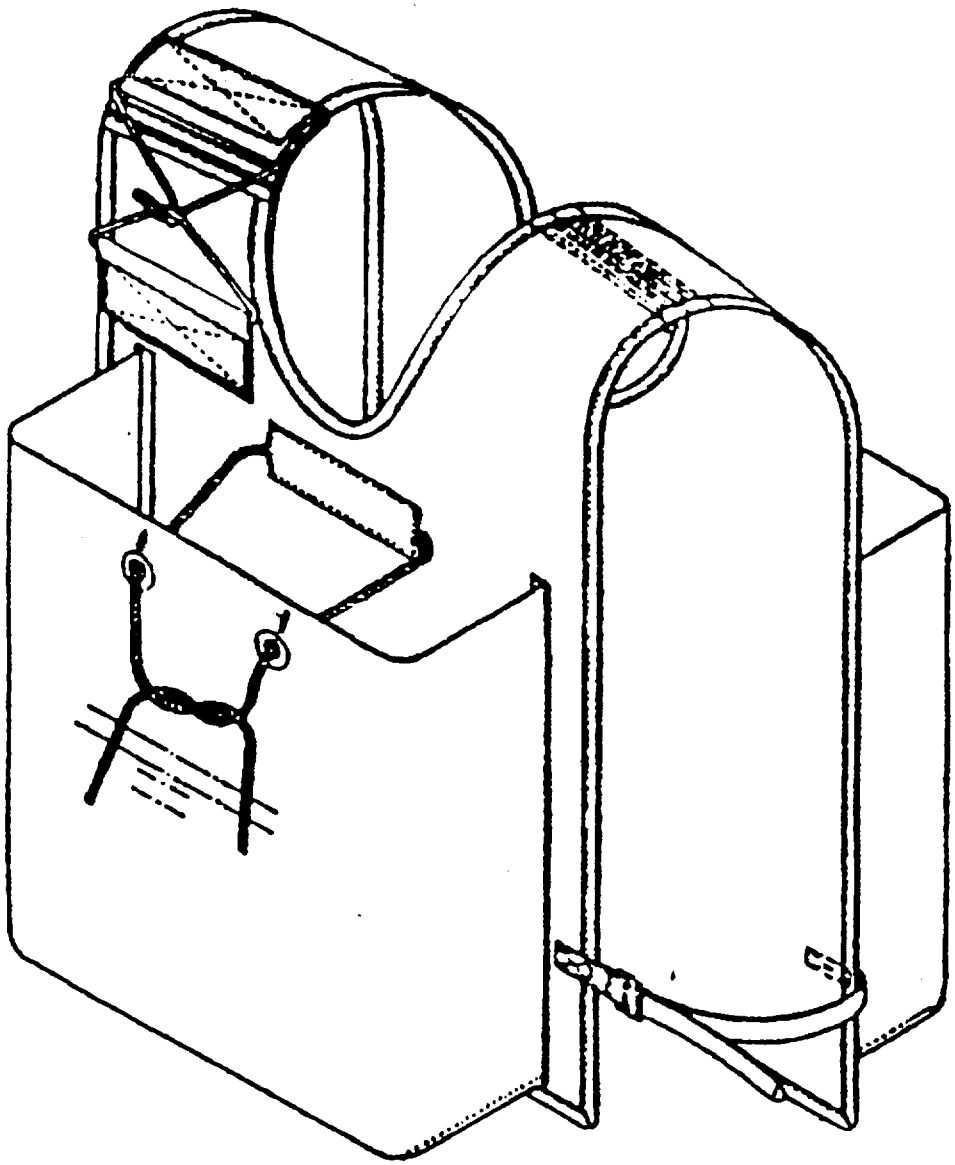


FIGURE 1: Vest, Ammunition Carrying

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-V-31001	25 FEB 1992

3. DOCUMENT TITLE

VEST, AMMUNITION CARRYING

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DEFENSE PERSONNEL SUPPORT CENTER	(1) Commercial (215) 737 8105	(2) AUTOVON 444-8105
c. ADDRESS (Include Zip Code) DPSC-FSSD 2800 South 20th Street Philadelphia, PA 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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